PROJECT: CONSOLIDATING EMPLOYEE DATA

datalab

You just got hired as the first and only data practitioner at a small business experiencing exponential growth. The company needs more structured processes, guidelines, and standards. Your first mission is to structure the human resources data. The data is currently scattered across teams and files and comes in various formats: Excel files, CSVs, JSON files...

You'll work with the following data in the `datasets` folder:

- **Office addresses**
  - Saved in `office_addresses.csv`.
  - If the value for office is `NaN`, then the employee is remote.
- **Employee addresses**
  - Saved on the first tab of `employee_information.xlsx`.
- **Employee emergency contacts**
  - Saved on the second tab of `employee_information.xlsx`; this tab is called `emergency_contacts`.
  - However, this sheet was edited at some point, and ***the headers were removed***! The HR manager let you know that they should be: `employee_id`, `last_name`, `first_name`, `emergency_contact`, `emergency_contact_number`, and `relationship`.
- **Employee roles, teams, and salaries**
  - This information has been exported from the company's human resources management system into a JSON file titled `employee_roles.json`. Here are the first few lines of that file:

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```
{"A2R5H9":
  {
    "title": "CEO",
    "monthly_salary": "$4500",
    "team": "Leadership"
  },
 ...
}
```

```python
import pandas as pd
# Start coding here...

# Read in office_addresses.csv
offices = pd.read_csv("datasets/office_addresses.csv")

# Read in employee_information.xlsx
addresses = pd.read_excel("datasets/employee_information.xlsx")

# Declare a list of new column names
emergency_contacts_header = ["employee_id", "last_name", "first_name",
                             "emergency_contact", "emergency_contact_number",
"relationship"]

# Read in employee_information.xlsx
emergency_contacts = pd.read_excel("datasets/employee_information.xlsx",
                                   sheet_name="emergency_contacts",
                                   header=None,
                                   names=emergency_contacts_header)

# Read in employee_roles.json
roles = pd.read_json("datasets/employee_roles.json", orient="index")

# Merge addresses with offices
employees = addresses.merge(offices, left_on="employee_country",
right_on="office_country", how="left")

# Merge employees with roles
employees = employees.merge(roles, left_on="employee_id", right_on=roles.index)

# Merge employees with emergency_contacts
employees = employees.merge(emergency_contacts, on="employee_id")

# Fill null values in office columns
for col in ["office", "office_country", "office_city", "office_street",
"office_street_number"]:
    employees[col].fillna("Remote", inplace=True)

# Create final columns
final_columns = ["employee_id", "employee_first_name", "employee_last_name",
"employee_country",
                 "employee_city", "employee_street", "employee_street_number",
                 "emergency_contact", "emergency_contact_number", "relationship",
                 "monthly_salary", "team", "title", "office", "office_country",
                 "office_city", "office_street", "office_street_number"]

# Subset for the required columns
```

```python
employees_final = employees[final_columns]

# Set employee_id as the index
employees_final.set_index("employee_id", inplace=True)
employees_final.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 4 entries, A2R5H9 to M1Z7U9
Data columns (total 17 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   employee_first_name        4 non-null      object
 1   employee_last_name         4 non-null      object
 2   employee_country           4 non-null      object
 3   employee_city              4 non-null      object
 4   employee_street            4 non-null      object
 5   employee_street_number     4 non-null      int64
 6   emergency_contact          4 non-null      object
 7   emergency_contact_number   4 non-null      object
 8   relationship               4 non-null      object
 9   monthly_salary             4 non-null      object
 10  team                       4 non-null      object
 11  title                      4 non-null      object
 12  office                     4 non-null      object
 13  office_country             4 non-null      object
 14  office_city                4 non-null      object
 15  office_street              4 non-null      object
 16  office_street_number       4 non-null      object
dtypes: int64(1), object(16)
memory usage: 576.0+ bytes
```